

SmartNPS360 JavaScript Bridge

Reference

Web ↔ native communication for the Android and iOS mobile app

Host: <https://smartnps360.com/>

Bridge handlers documented: 18

App version: 1.0.0+55

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Overview

The SmartNPS360 mobile app loads the web app inside an in-app WebView. JavaScript on the page can call native features (GPS, files, push, auth, clock-in) through a bridge. The native app can also push status updates and geolocation fixes back into the page.

All bridge calls use `window.flutter_inappwebview.callHandler(handlerName, payload)` and return a Promise. Injected scripts also expose higher-level helpers on `window.SmartNPS360` and `window.SmartNPSNativeAuth`.

NATIVE APP DETECTION

- User-Agent contains `SmartNPS360App`
- `window.SmartNPS360.isNativeApp()` returns `true` in the mobile shell

TRUSTED ORIGINS

Most handlers only accept calls when the current page host is `smartnps360.com`, `www.smartnps360.com`, or a subdomain of `smartnps360.com`. Untrusted callers receive:

```
{
  "ok": false,
  "error": { "code": "untrusted_origin", "message": "Untrusted origin" }
}
```

COMMON RESPONSE ENVELOPE

```
{ "ok": true, /* payload fields */ }
{ "ok": false, "error": { "code": "string", "message": "string" } }
```

INJECTED SCRIPTS (AT PAGE LOAD)

Script	Purpose
SmartNPS360 bridge	Auth helpers, push toggles, clock-in gate, theme, legacy location bridge
Geolocation override	Replaces <code>navigator.geolocation</code> with native GPS
iOS popover fix	Layout fix for notification UI on iOS (internal)

For web developers: Use `bindClockInGate` or `ensureCanClockIn()` before clock-in. Do not submit attendance directly on button click without the native background-location gate. See the clock-in integration example in `web_integration/clock_in_page.example.js`.

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1. Auth & Session

Web → Native

authEvent

WEB ENTRY POINTS

- SmartNPSNativeAuth.login(user, session)
- SmartNPSNativeAuth.setSession(session)
- SmartNPSNativeAuth.logout()

REQUEST

Field	Type	Required	Description
action or type	string	Yes	login, signup, register, session, or logout
user or profile	object	For login/signup	Officer profile object
session, auth, or tokens	object	For login/session	Token bundle (access token, refresh token, expiry fields)

EXAMPLE — LOGIN

```
{
  "action": "login",
  "user": { "id": 42, "name": "Officer Smith" },
  "session": { "accessToken": "eyJ...", "refreshToken": "..."}
}
```

RESPONSES

Action	Success	Failure
login / signup	{ "ok": true, "action": "login" }	invalid_args if user missing
session	{ "ok": true, "action": "session" }	invalid_args if session missing

Action	Success	Failure
logout	{ "ok": true, "action": "logout" }	logout_blocked_on_duty if officer is still on duty

NATIVE SIDE EFFECTS

On login/session: stores tokens, marks officer logged in, starts duty heartbeat, syncs push token. On logout: clears session, stops tracking, removes push token (blocked while on duty).

Web → Native loginWithSanctum

DESCRIPTION

Native Sanctum login. Also auto-triggered when a login form with employee_no and password fields is submitted.

REQUEST

```
{ "username": "EMP001", "password": "secret" }
```

username is sent to the API as employee_no.

RESPONSE

```
{ "ok": true, "hasToken": true }
{ "ok": false, "error": { "code": "request_failed", "message": "Sanctum login failed" } }
```

2. Location & Clock-in

Web → Native `getCurrentLocation`

WEB ENTRY POINTS

- `Direct callHandler('getCurrentLocation', payload)`
- `navigator.geolocation.getCurrentPosition` (injected override)
- `SmartNPSBridge.postMessage / SmartNPSAndroid.postMessage` with `request_current_location`

REQUEST OPTIONS

Field	Type	Default	Description
<code>timeout_ms</code>	number	12000	Clamped 5000–45000
<code>required_accuracy_meters</code>	number	50	Clamped 5–500
<code>for_clock_in</code> or <code>purpose</code>	bool / string	false	Enables clock-in disclosure and mock-location checks
<code>allow_foreground_only</code>	boolean	false	Skips background-location gate

SUCCESS RESPONSE

```
{
  "ok": true,
  "location": {
    "latitude": 24.8607,
    "longitude": 67.0011,
    "accuracy": 12.5,
    "altitudeAccuracy": null,
    "timestampMs": 1752494400000,
    "timestamp": "2026-07-14T12:00:00.000Z",
    "altitude": 45.0,
    "speed": 0.0,
    "speedAccuracy": null,
    "heading": 0.0,
    "headingAccuracy": null,
    "isMocked": false,
    "isSimulatedBySoftware": false,
  }
}
```

```

    "floor": null,
    "acceptedFromLiveStream": true,
    "isFreshLiveLocation": true,
    "isCachedLocation": false,
    "ageMsWhenAccepted": 0,
    "maxAllowedAccuracyMeters": 50,
    "timeoutMs": 12000,
    "effectiveMaxAllowedAccuracyMeters": 50,
    "degradedAccuracyAccepted": false
  }
}

```

LOCATION OBJECT FIELDS

Field	Type	Description
latitude	number	GPS latitude
longitude	number	GPS longitude
accuracy	number	Horizontal accuracy in meters
altitudeAccuracy	number null	Altitude accuracy when available
timestampMs	number	Fix time in milliseconds
timestamp	string	Fix time ISO 8601
altitude	number	Altitude in meters
speed	number	Speed in m/s
speedAccuracy	number null	Speed accuracy when available
heading	number	Heading in degrees
headingAccuracy	number null	Heading accuracy when available
isMocked	boolean	Whether the fix is mocked
isSimulatedBySoftware	boolean	Whether the fix is software-simulated
floor	number null	Floor level when available
acceptedFromLiveStream	boolean	True when accepted from live GPS stream

Field	Type	Description
isFreshLiveLocation	boolean	True for a fresh live fix
isCachedLocation	boolean	True when a cached fix was used
ageMsWhenAccepted	number	Age of the fix when accepted (ms)
maxAllowedAccuracyMeters	number	Requested accuracy ceiling
timeoutMs	number	Request timeout used
effectiveMaxAllowedAccuracyMeters	number	Effective accuracy ceiling applied
degradedAccuracyAccepted	boolean	True only if a degraded fix was accepted (currently always false)

ERROR RESPONSE (ACCURACY TIMEOUT)

```
{
  "ok": false,
  "error": {
    "code": "fresh_location_unavailable",
    "message": "Unable to get a GPS location with accuracy <= 50 meters."
  },
  "bestAccuracySeenMeters": 87.5
}
```

COMMON ERROR CODES

untrusted_origin, clock_in_gate_in_progress, location_services_disabled, background_location_required, location_precise, disclosure_required, permission_denied, permission_denied_forever, mock_location, fresh_location_unavailable, get_location_failed

Web ↔ Native startLocationWatch

DESCRIPTION

Starts a native GPS stream. Native pushes updates via `__smartnps_native_geo_emit`.

REQUEST

```
{
  "watchId": 900001,
  "options": {
    "required_accuracy_meters": 50,
    "timeout_ms": 12000,
    "interval_ms": 1000,
    "for_clock_in": true,
    "allow_foreground_only": false
  }
}
```

RESPONSE

```
{ "ok": true, "watchId": 900001, "intervalMs": 1000 }
```

[Web → Native](#) clearLocationWatch

REQUEST

```
{ "watchId": 900001 }
```

RESPONSE

```
{ "ok": true, "watchId": 900001 }
```

[Web → Native](#) prepareClockIn

WEB ENTRY POINT

SmartNPS360.ensureCanClockIn()

DESCRIPTION

Runs the store-safe clock-in gate: disclosure, permissions, background location readiness.

RESPONSE — READY

```
{ "ok": true, "canClockIn": true, "prepareInFlight": false }
```

RESPONSE — BLOCKED

```
{
  "ok": true,
  "canClockIn": false,
  "prepareInFlight": false,
  "reason": "background_location_required",
  "title": "Background location required for shift attendance",
  "message": "Enable background location (Always / Allow all the time) ..."
}
```

BLOCKED REASON CODES

clock_in_in_progress, location_services_disabled, background_location_required, location_precise, disclosure_cancelled / cancelled variants, mock_location

Clock-in requires background location **and** precise location. When precise location is off, reason is location_precise.

[Web → Native](#) getBackgroundLocationStatus

WEB ENTRY POINT

SmartNPS360.getBackgroundLocationStatus()

RESPONSE

```
{
  "ok": true,
  "backgroundReady": false,
  "phase": "foregroundOnly",
  "deniedReason": "background_location_required",
  "disclosureAccepted": true,
  "serviceEnabled": true,
  "canClockIn": false,
  "prepareInFlight": false,
  "title": "Background location required...",
  "message": "Enable Always...",
  "platform": "android"
}
```

phase: none | foregroundOnly | backgroundReady

RESPONSE FIELDS

Field	Type	Description
backgroundReady	boolean	True when Always / Allow all the time is granted
phase	string	Permission phase (see above)
deniedReason	string null	e.g. background_location_required, location_precise, location_services_disabled
disclosureAccepted	boolean	Whether location disclosure was accepted
serviceEnabled	boolean	Whether OS location services are on
canClockIn	boolean	True when background + precise + services ready and gate not in flight
prepareInFlight	boolean	True while prepareClockIn is running
title	string null	Blocked title when canClockIn is false
message	string null	Blocked message when canClockIn is false
platform	string	android or ios

3. Push Notifications

Web → Native `getPushNotificationToken`

RESPONSE

```
{ "ok": true, "token": "fcm_token_string_or_null" }
```

Web → Native `getPushNotificationStatus`

WEB ENTRY POINTS

`SmartNPS360.getPushNotificationStatus(), syncPushNotifications()`

RESPONSE

```
{  
  "ok": true,  
  "enabled": true,  
  "permissionGranted": true,  
  "hasToken": true  
}
```

Web → Native `setPushNotificationsEnabled`

WEB ENTRY POINT

`SmartNPS360.setPushNotificationsEnabled(enabled)`

REQUEST

`true / false` or `{ "enabled": true }`

RESPONSE

Same shape as `getPushNotificationStatus`. May include `"unchanged": true`.

4. Files & Sharing

Web → Native pickFile

RESPONSE — SELECTED

```
{
  "ok": true,
  "canceled": false,
  "file": { "path": "/path/to/file.pdf", "name": "file.pdf", "size": 12345, "extension": "pdf" }
}
```

RESPONSE — CANCELED

```
{ "ok": true, "canceled": true }
```

Web → Native pickImage

RESPONSE

```
{ "ok": true, "canceled": false, "image": { "path": "...", "name": "..." } }
```

Web → Native takePhoto

RESPONSE

```
{ "ok": true, "canceled": false, "photo": { "path": "...", "name": "..." } }
```

Web → Native downloadFile

REQUEST

URL string or { "url": "https://...", "filename": "report.pdf" }

RESPONSE

```
{ "ok": true, "file": { "path": "/app/documents/downloads/report.pdf", "name": "report.pdf" } }
```

Web → Native shareContent

REQUEST

Text string or { "text": "...", "subject": "..." }

RESPONSE

```
{ "ok": true }
```

5. Device & Navigation

Web → Native `getDeviceInfo`

RESPONSE (ANDROID EXAMPLE)

```
{
  "ok": true,
  "platform": "android",
  "device": { "brand": "Google", "manufacturer": "Google", "model": "Pixel 8", "sdkInt":
34 }
}
```

RESPONSE (IOS EXAMPLE)

```
{
  "ok": true,
  "platform": "ios",
  "device": { "name": "iPhone", "model": "iPhone", "systemName": "iOS", "systemVersion":
"18.0" }
}
```

Web → Native `openExternalUrl`

REQUEST

URL string or { "url": "https://..." }

RESPONSE

```
{ "ok": true, "launched": true }
```

NAVIGATION (NATIVE, NOT A HANDLER)

Behavior	Description
Internal URLs	Loaded inside the WebView
External http(s) links	Opened in the system browser

Behavior	Description
Logout routes (/officer/logout, */logout)	Trigger native logout if officer is not on duty
Login routes	Pause native session and stop duty heartbeat
Push notification tap	Native app loads deep-link URL directly

6. Theme

Web → Native

themeChanged

WEB ENTRY POINTS

- `SmartNPS360.notifyTheme('dark')` or `notifyTheme('light')`
- Auto-detected via `DOM MutationObserver` on page theme classes

REQUEST

Boolean or { "isDark": true, "theme": "dark", "mode": "dark" }

RESPONSE

```
{ "ok": true, "isDark": true }
```

Updates native status bar and navigation bar colors.

7. Native → Web Events & Callbacks

CUSTOM DOM EVENTS

Event	Detail	When fired
smartnps360:background-location	getBackgroundLocationStatus result	Permission changes, page load
smartnps360:push-notifications	Push status object	Page load, app resume
smartnps360:push-token	{ "token": "fcm..." }	iOS after login / token sync

CALLBACK HOOKS ON WINDOW.SMARTNPS360

Callback	Argument
onBackgroundLocationStatusChanged(status)	Background location status object
onPushNotificationStatusChanged(status)	Push status object
onPushTokenReady(token)	FCM token string (iOS)

GEOLOCATION WATCH CALLBACKS

Function	Payload
window.__smartnps_native_geo_emit(watchId, position)	Standard Geolocation position object with native metadata
window.__smartnps_native_geo_error(watchId, err)	{ "code": 2, "message": "..." }

WATCH POSITION PAYLOAD

```
{
  "coords": {
    "latitude": 24.8607,
    "longitude": 67.0011,
    "accuracy": 12.5,
```

```
    "altitude": 45.0,  
    "altitudeAccuracy": null,  
    "heading": 0.0,  
    "speed": 0.0  
  },  
  "timestamp": 1752494400000,  
  "nativeSource": true,  
  "provider": "flutter_geolocator",  
  "is_mocked": false,  
  "is_simulated_by_software": false,  
  "accepted_from_live_stream": true,  
  "max_allowed_accuracy_meters": 50,  
  "timeout_ms": 12000  
}
```

8. Web Helper API (window.SmartNPS360)

Injected at document start. All methods are no-ops or return rejected promises in a normal browser.

Method	Handler / behavior	Category
isNativeApp()	User-Agent check	Detection
notifyTheme(mode)	themeChanged	Theme
getBackgroundLocationStatus()	getBackgroundLocationStatus	Clock-in
ensureCanClockIn()	prepareClockIn	Clock-in
bindClockInGate(button, onAllowed, onBlocked)	Gate + status listener	Clock-in
getPushNotificationStatus()	getPushNotificationStatus	Push
setPushNotificationsEnabled(enabled)	setPushNotificationsEnabled	Push
syncPushNotifications()	Get status + broadcast	Push
broadcastPushNotificationStatus(status)	DOM event + UI sync	Push
bindPushNotificationToggle(input)	Auto-binds [data-smartnps-push-toggle]	Push
bindAlpinePushNotificationToggle(button)	Alpine.js toggle helper	Push

WINDOW.SMARTNPSNATIVEAUTH

Method	Sends to authEvent
login(user, session)	{ action: "login", user, session }
setSession(session)	{ action: "session", session }
logout()	{ action: "logout" }

9. Geolocation Override

Browser geolocation is disabled in WebView settings. The injected script replaces `navigator.geolocation` with native GPS.

Browser API	Native behavior
<code>getCurrentPosition(success, error, options)</code>	Runs clock-in gate (unless <code>allow_foreground_only</code>), then <code>getCurrentLocation</code>
<code>watchPosition(success, error, options)</code>	<code>startLocationWatch</code> + <code>__smartnps_native_geo_emit</code>
<code>clearWatch(id)</code>	<code>clearLocationWatch</code>

WEB OPTIONS → NATIVE OPTIONS

Web option	Native option
<code>enableHighAccuracy: false</code>	<code>required_accuracy_meters: 100</code>
<code>timeout</code>	<code>timeout_ms</code> (5000–45000)
<code>required_accuracy_meters</code>	Passed through (5–500)
<code>allow_foreground_only: true</code>	Skips clock-in gate
Clock-in gate unlocked	Sets <code>for_clock_in: true</code>

10. Legacy Location Bridge

For older web integrations that do not use the geolocation override.

Web → Native

SmartNPSBridge.postMessage / SmartNPSAndroid.postMessage

REQUEST

```
{ "action": "request_current_location" }
```

Native → Web

SmartNPSWeb.receiveLocation / receiveLocationError

SUCCESS CALLBACK PAYLOAD

```
{
  "latitude": 24.8607,
  "longitude": 67.0011,
  "accuracy": 12.5,
  "provider": "flutter_geolocator",
  "timestamp": 1752494400000,
  "timestamp_iso": "2026-07-14T12:00:00.000Z",
  "altitude": 45.0,
  "speed": 0.0,
  "heading": 0.0,
  "is_mocked": false,
  "is_simulated_by_software": false,
  "accepted_from_live_stream": true,
  "max_allowed_accuracy_meters": 50,
  "timeout_ms": 12000
}
```

ERROR CALLBACK PAYLOAD

```
{ "code": "NATIVE_GPS_FAILED", "message": "..." }
```

SmartNPS360 JavaScript Bridge Reference — mobile app 1.0.0+55 (updated 28 July 2026).

Handler registration summary: pickFile, pickImage, takePhoto, getCurrentLocation, startLocationWatch, clearLocationWatch, getDeviceInfo, openExternalUrl, shareContent, downloadFile, getPushNotificationToken, getPushNotificationStatus, setPushNotificationsEnabled, getBackgroundLocationStatus, prepareClockIn, themeChanged, loginWithSanctum, authEvent.